

Applying Bantu RGs in South Africa and beyond

Laurette Marais, GFSS 2025



Overview

- Background
- Ngiyaqonda! Foundation Phase Literacy
- Wikidata for the people of Africa
- Data annotation and augmentation
- Universal Dependencies treebanks with gf2ud
- What's next?

Background

Where we were 2 years ago:

We want to use GF to solve problems

- Arising from being a **multilingual** society
- Arising from being an **historically unequal** society
- Arising from dealing with **low-resourced** languages
- Arising from dealing with **morphologically rich** languages

What do we have going for us?

- Very close structural correlation between most of the languages! 😊
- GF

Background

We split our efforts into two main themes:

- GF for applications
- GF for language and NLP resources

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- GF for applications
 - Literacy and language learning
 - Natural language generation from structured data
- GF for language and NLP resources

Background

We split our efforts into two main themes:

- GF for applications
 - Literacy and language learning
 - Natural language generation from structured data
- GF for language and NLP resources
 - Parsing
 - Morphological annotation
 - Syntactic annotation
 - Data augmentation

Ngiyaqonda! Foundation Phase Literacy

What is the problem?

- 8 out of 10 Grade 4 learners in South Africa cannot read for basic meaning in their home language.

Why is this a problem?

- Historically unequal society
- Multilingual society
- Low-resourced languages
- Morphologically rich languages

Historically unequal, multilingual society

Typical problems of a post-liberation African democracy

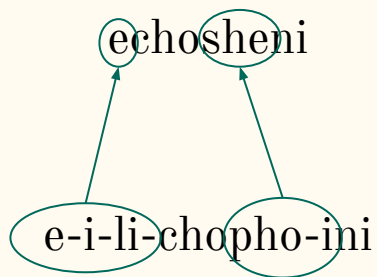
- Political agenda governed by ruling party factional politics
- Very little political will for solving problems like literacy (especially with 11 official languages)
- Infrastructure collapse and worsening economic situation
 - forces problems like literacy to the background
 - reduces the capacity of the state (and private sector) to fund literacy initiatives
- Attention and resources divided by 11
- Children are taught in their home language in Grades 1-3, and typically in English in Grades 4-12.

Low-resourced languages

It's low-resourcedness all the way down!

- Language and literature content
- Low mother-tongue literacy
- Few experts
- Very little data - even less quality data
- No useable LLMs whatsoever...

Morphologically rich languages



at the (magnetic) pole

Agglutinating morphology

Nguni family exhibits extensive morphophonological alternation

Ngiyaqonda! Foundation Phase Literacy

How can GF power a solution?

- Enable natural language generation using age appropriate vocabulary and syntactic complexity
- Multilingual story-based and vocabulary-based application grammars aimed at Gr3 learners (8-9 years old)
- Utilise TTS (text-to-speech) and APED (automatic pronunciation error detection) to provide a rich learning experience

Ngiyaqonda! Foundation Phase Literacy



Ngiyaqonda! Foundation Phase Literacy

Current status

- Waiting for follow-up funding
- Proposed next steps
 - Add support for isiXhosa, Setswana and Tshivenda
 - (Develop RGs for Setswana and Tshivenda)
 - Extend application grammars to support ages 6-9
 - Develop structured lexical resources for phonetically informed NLG
 - Release app on the Playstore

Wikidata for the people of Africa

Wikidata is “a free and open knowledge base that can be read and edited by both humans and machines”.

- Editing, consuming, browsing, and reusing the data is fully multilingual
- Editing in any language is possible and encouraged

Wikidata for the people of Africa

Entities (and properties) have natural language *labels* and *descriptions*.

Botswana (Q963) 🔒

[Item](#) [Discussion](#) Read View history ☆

sovereign state in Southern Africa
bw | Republic of Botswana | Lefatshe la Botswana | BOT

[In more languages](#)

Language	Label	Description	Also known as
English	Botswana	sovereign state in Southern Africa	bw Republic of Botswana Lefatshe la Botswana BOT
Afrikaans	Botswana	Landingslote land in Suider-Afrika	
default for all languages	No label defined	–	BW BWA BOT
Zulu	IButswana	izwe elingenalo ulwandle elise-Southern Africa	

[All entered languages](#)

More on Wednesday...

Data annotation and augmentation

Morphological segmentation

- Data scarcity due to low-resourced status
- Data sparsity due to agglutinating morphology

“ngizombophela” (*I will tie them*)

“echosheni” (*at the pole*)

- Canonical segmentation
 - ngi-zo-mu-bophel-a
 - e-i-li-chopho-ini
- Surface segmentation
 - ngi-zo-m-bophel-a
 - e-cho-sh-e-ni

Data annotation and augmentation

“ngizombophela” (*I will tie them*)

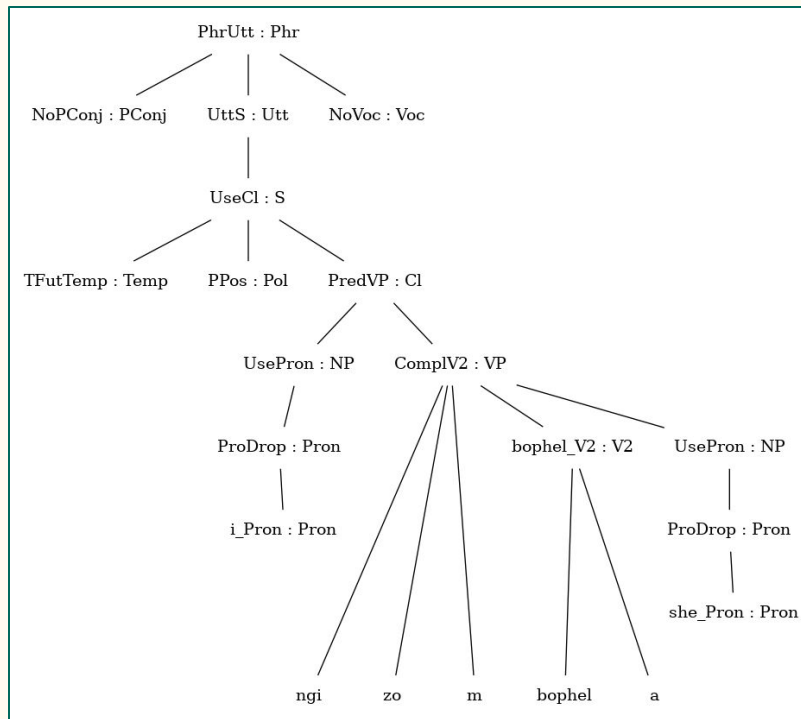
- Canonical segmentation
 - ngi-zo-mu-bophel-a
 - Smaller vocabulary
 - No morphophonological alternation
- Surface segmentation
 - ngi-zo-m-bophel-a
 - Trivially reversible: can be used for generation

Data annotation and augmentation

“ngizombophela” (*I will tie them*)

- Surface segmentation
 - ngi-zo-m-bophel-a

The isiZulu RG provides a “natural” level of segmentation. What about finer grained or more coarse grained?



Data annotation and augmentation

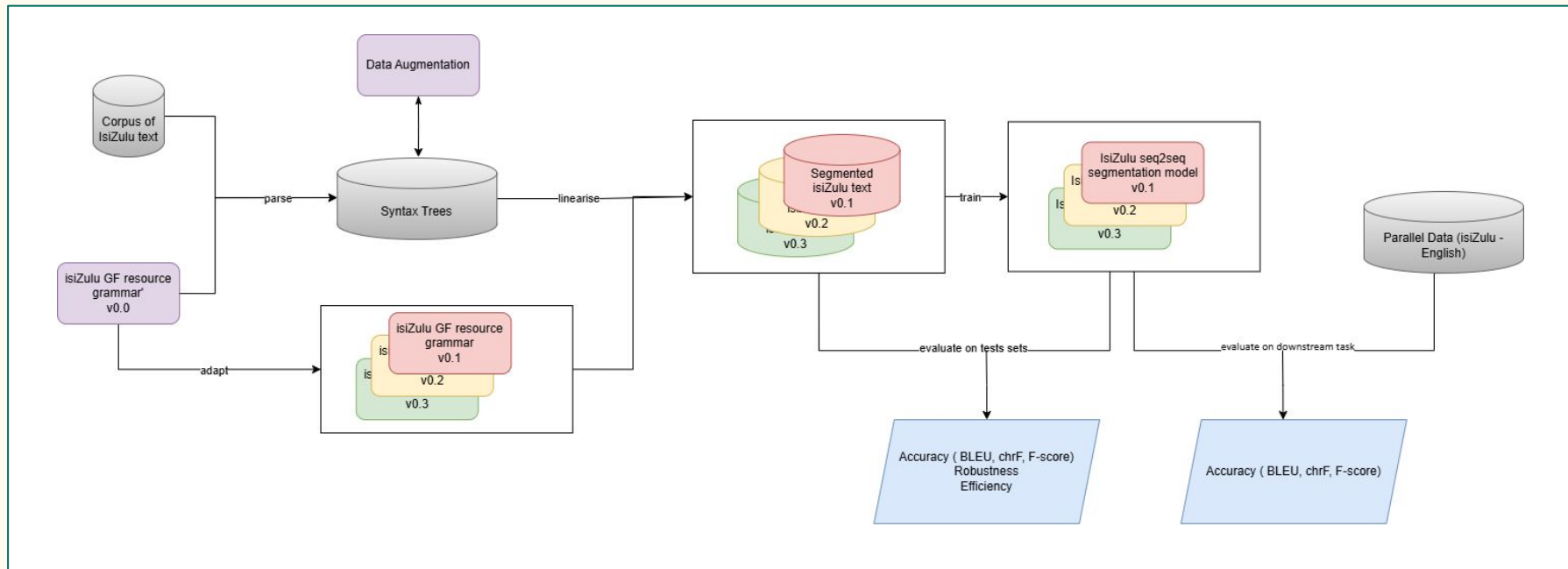
Annotation via different RG versions

1. “Natural” segmentation
2. Fine grained
 - Insertion of more BIND tokens
3. Coarse grained
 - Replace some BIND tokens with special tokens for selective gluing

Augmentation

- Shuffle nouns, pronouns, verbs
- Changes to number, polarity, tense

Data annotation and augmentation



Mkhwanazi, Sthembiso. *Investigating supervised surface segmentation of isiZulu text using synthetic data generation*. Thesis. Potchefstroom: North-West University, 2025.

Data annotation and augmentation

Evaluating the effect of segmentation on isiZulu-to-English translation.

Model	BLEU	chrF
Segmenter One	0.227	0.460
Segmenter Two	0.235	0.461
Segmenter Three	0.224	0.459
No Segmentation	0.188	0.420

In-domain test set

Model	BLEU	chrF
Segmenter One	0.1056	0.4021
Segmenter Two	0.1019	0.3973
Segmenter Three	0.0891	0.3767
No Segmentation	0.0764	0.3542

Out-of-domain test set

Data annotation

What about morphological annotation?

- In order to streamline bootstrapping, we already have ALL strings in the ResZul module
- Decorate the strings with ZulMorph tags:
 - TEMP_PREF_FUT : Str = "zo" ;
 - TEMP_PREF_FUT : Str = "zo [Fut] " ;

Morphological annotation

Decorate strings with morphological tags

```
-- with
instrPref : RInit => Str = table {
  RU => "ngo" ;
  RI => "nge" ;
  RO => "ngo" ;
  _  => "nga"
} ;
```



```
-- with
instrPref : RInit => Str = table {
  RU => "ngo[AdvPre]" ;
  RI => "nge[AdvPre]" ;
  RO => "ngo[AdvPre]" ;
  _  => "nga[AdvPre]"
} ;
```

Morphological annotation

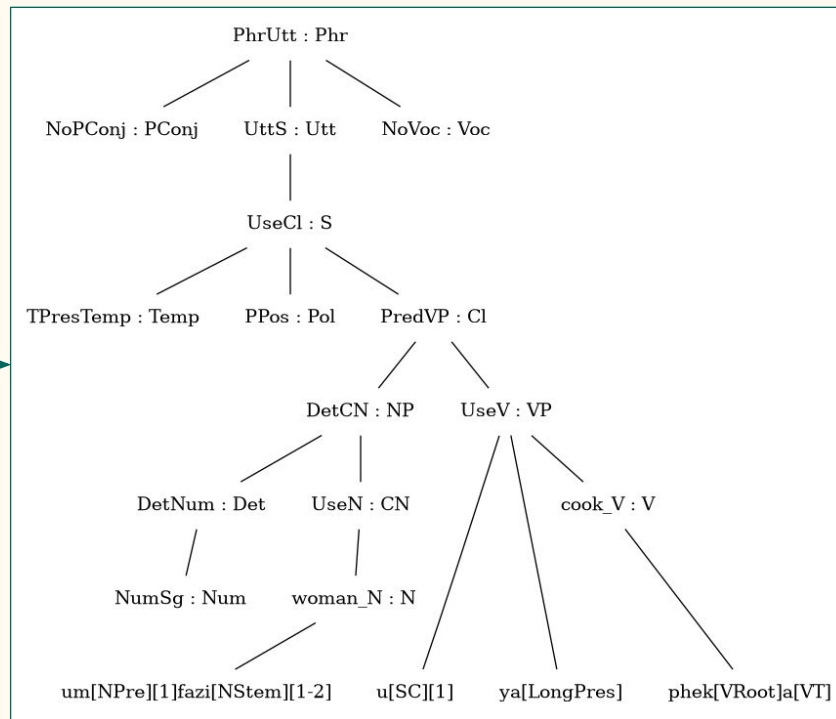
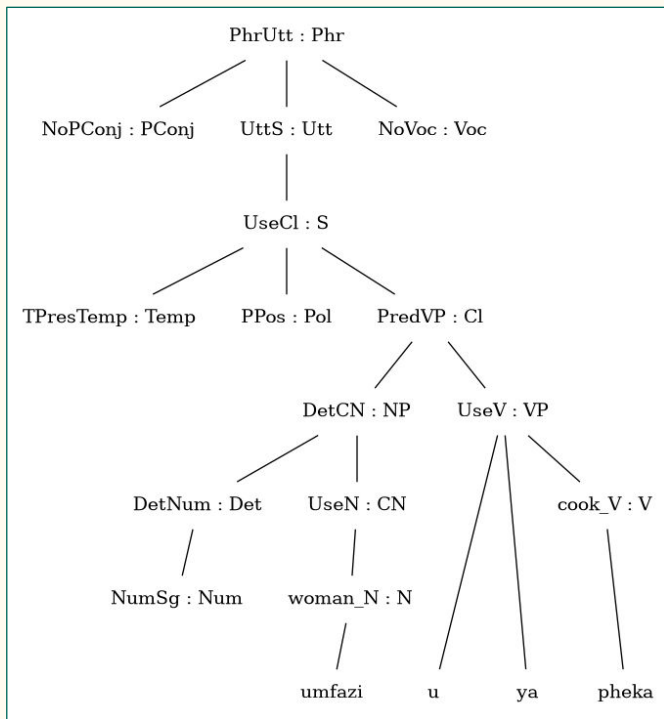
Adapt opers where necessary

```
prefix_nasal : Str -> Str = \root -> case root of {  
  "ph"+x => "mp" + x ;  
  "Ph"+x => "mP" + x ;  
  "bh"+x => "mb" + x ;  
  "Bh"+x => "mB" + x ;  
  -- ...  
}
```

```
prefix_nasal : Str -> ClassGender -> Number -> Str = \root,classgender,number -> let  
  class_tag = noun_prefix_tag classgender number ;  
in  
  case root of {  
    "ph"+x => "m[NPre]"+class_tag+"p" + x ;  
    "Ph"+x => "m[NPre]"+class_tag+"P" + x ;  
    "bh"+x => "m[NPre]"+class_tag+"b" + x ;  
    "Bh"+x => "m[NPre]"+class_tag+"B" + x ;  
    -- ...  
  }
```

```
noun_prefix_tag : ClassGender -> Number -> Str = \c,n -> case <c,n> of {  
  <C1_2,Sg> => "[1]" ;  
  <C1_2,Pl> => "[2]" ;  
  <C1a_2a,Sg> => "[1a]" ;  
  <C1a_2a,Pl> => "[2a]" ;  
  <C3_4,Sg> => "[3]" ;  
  <C3_4,Pl> => "[4]" ;  
  <C5_6,Sg> => "[5]" ;  
  <C5_6,Pl> => "[6]" ;  
  <C7_8,Sg> => "[7]" ;  
  <C7_8,Pl> => "[8]" ;  
  <C9_10,Sg> => "[9]" ;  
  <C9_10,Pl> => "[10]" ;  
  <C11_10,Sg> => "[11]" ;  
  <C11_10,Pl> => "[10]" ;  
  <C9_6,Sg> => "[9]" ;  
  <C9_6,Pl> => "[6]" ;  
  <C14,_> => "[14]" ;  
  <C15,_> => "[15]" ;  
  <C17,_> => "[17]" ;  
}
```


Morphological annotation



Universal Dependencies treebanks with gf2ud

Automatically convert GF treebanks to UD treebanks

1. Obtain GF treebank
2. Define abstract annotations
3. Define concrete annotations
4. Obtain UD treebank!

The UD project contains one Southern Bantu treebank of 25 sentences for Setswana.

gf2ud

Ranta, Aarne, and Prasanth Kolachina. "From universal dependencies to abstract syntax." *Proceedings of the NoDaLiDa 2017 Workshop on Universal Dependencies (UDW 2017)*. 2017.

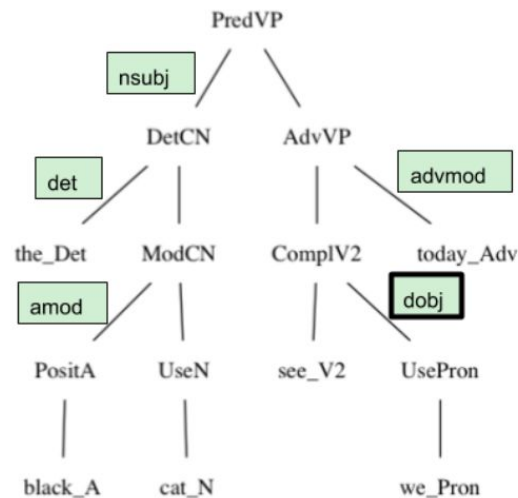
abstract syntax

PredVP : NP -> VP -> C1
ComplV2 : V2 -> NP -> VP
AdvVP : VP -> Adv -> VP
DetCN : Det -> CN -> NP
ModCN : AP -> CN -> CN
UseN : N -> CN
UsePron : Pron -> NP
PositA : A -> AP

cat A
cat Adv
cat Det
cat N
cat Pron
cat V2

dependency configuration

nsubj head
head dobj
head advmod
det head
amod head
head
head
Head



gf2ud

abstract syntax

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head

head

Head

ADJ

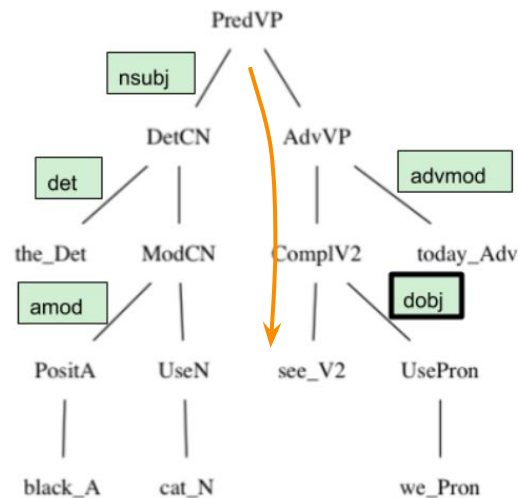
ADV

DET

NOUN

PRON

VERB



gf2ud

abstract syntax

PredVP : NP -> VP -> C1

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DetCN : Det -> CN -> NP

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Head

ADJ

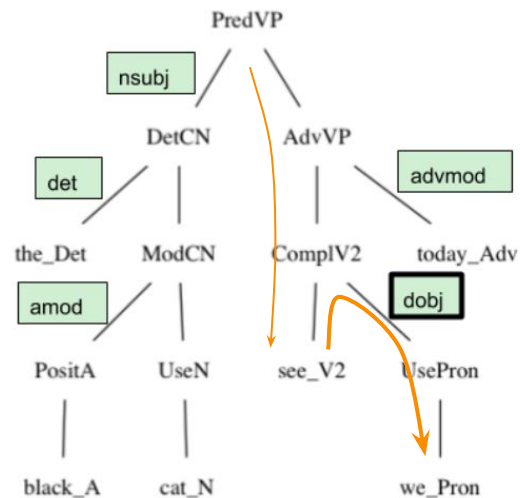
ADV

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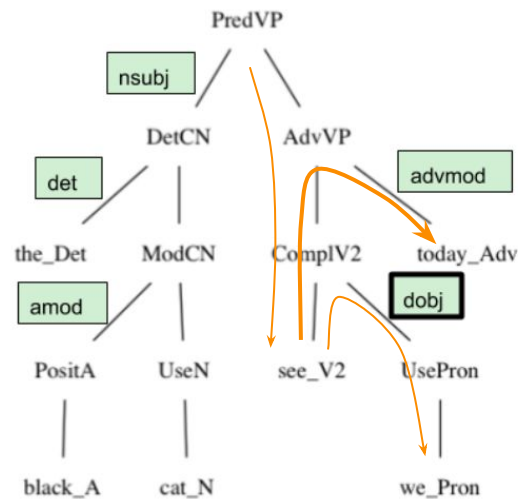
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cat A

cat Adv

cat Det

cat N

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cat V2

dependency configuration

nsubj head

head dobj

head advmod

det head

amod head

head

head

Head

ADJ

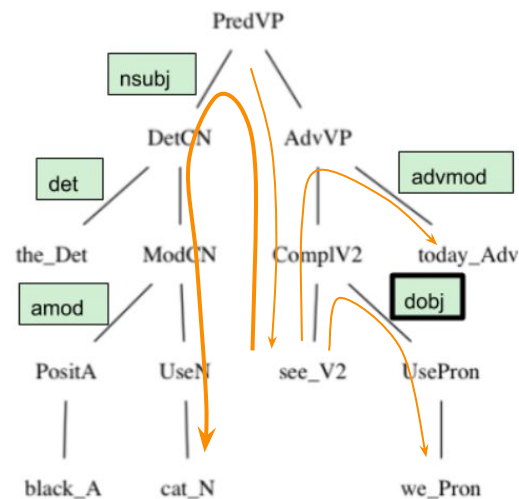
ADV

DET

NOUN

PRON

VERB



gf2ud

dependency configuration

```
nsubj    head
```

head dobj

head advmod

det head

amod head

head

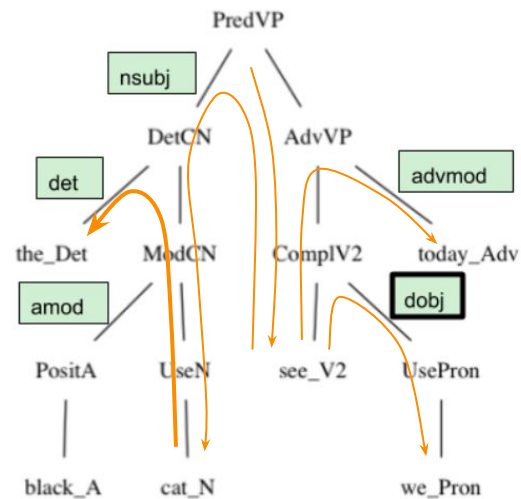
head

HeadADJADVDET

NOUN

PRON

VERB



gf2ud

abstract syntax

PredVP : NP -> VP -> C1

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cat A

cat Adv

cat Det

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cat V2

dependency configuration

nsubj head

head dobj

head advmod

det head

amod head

head

head

Head

ADJ

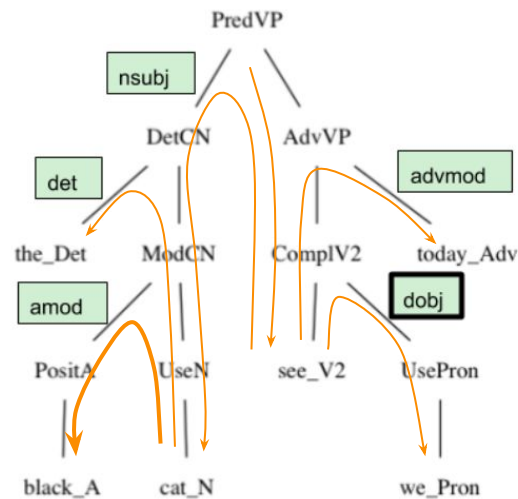
ADV

DET

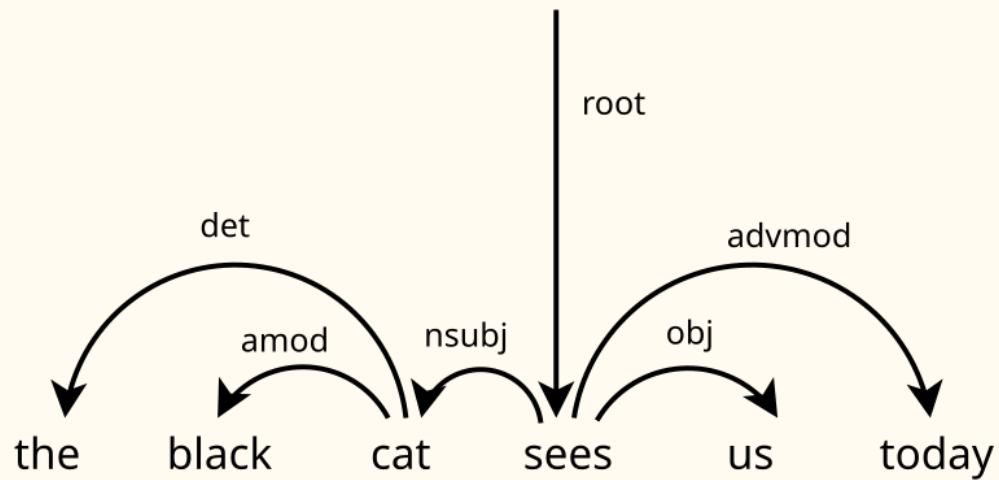
NOUN

PRON

VERB



gf2ud



gf2ud for isiZulu

Define abstract annotations

```
#fun PredVP nsubj head -- NP -> VP -> Cl
```

```
#fun RelNP head acl:relcl -- NP -> RS -> NP
```

```
#fun PronPostdetNP head det -- Pron -> Postdet -> NP
```

gf2ud for isiZulu

Define concrete annotations?

```
#word is be Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin
```

```
#lemma UseAP,UseAdv,UseNP be Cop cop head
```

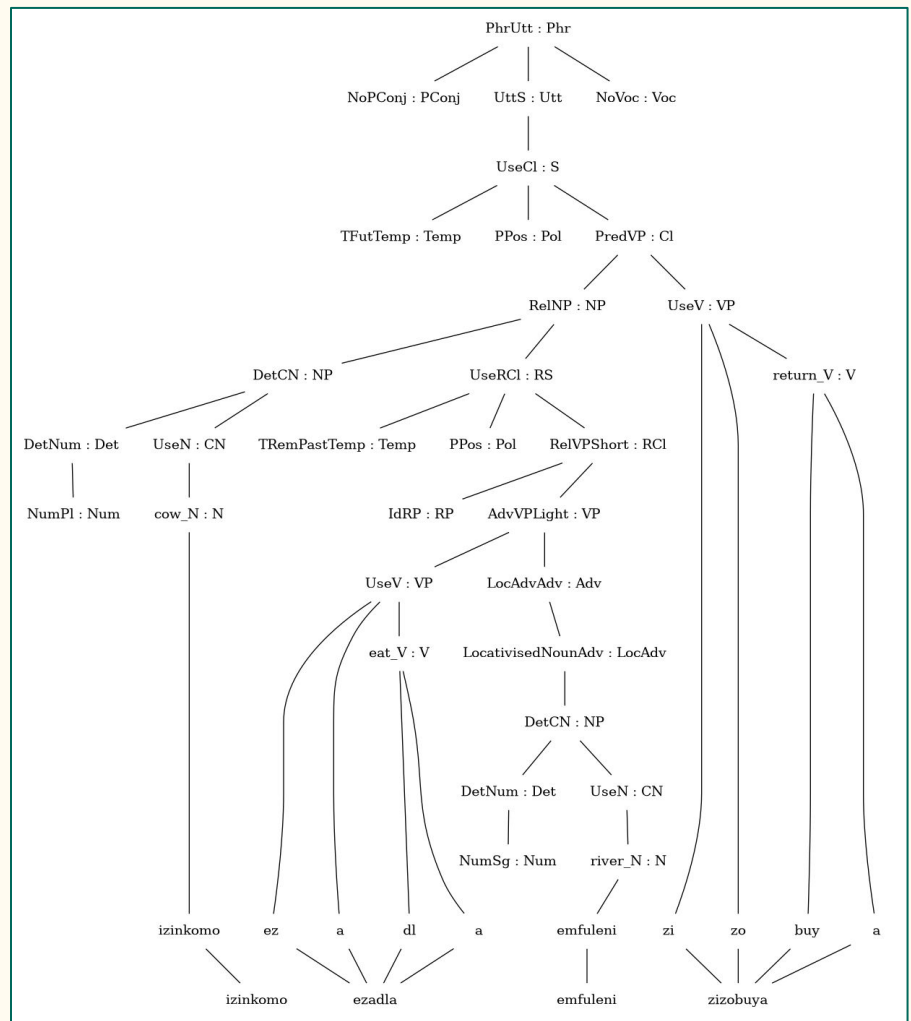
```
#word ngi SC ...
```

```
#word ngi OC ... ?
```

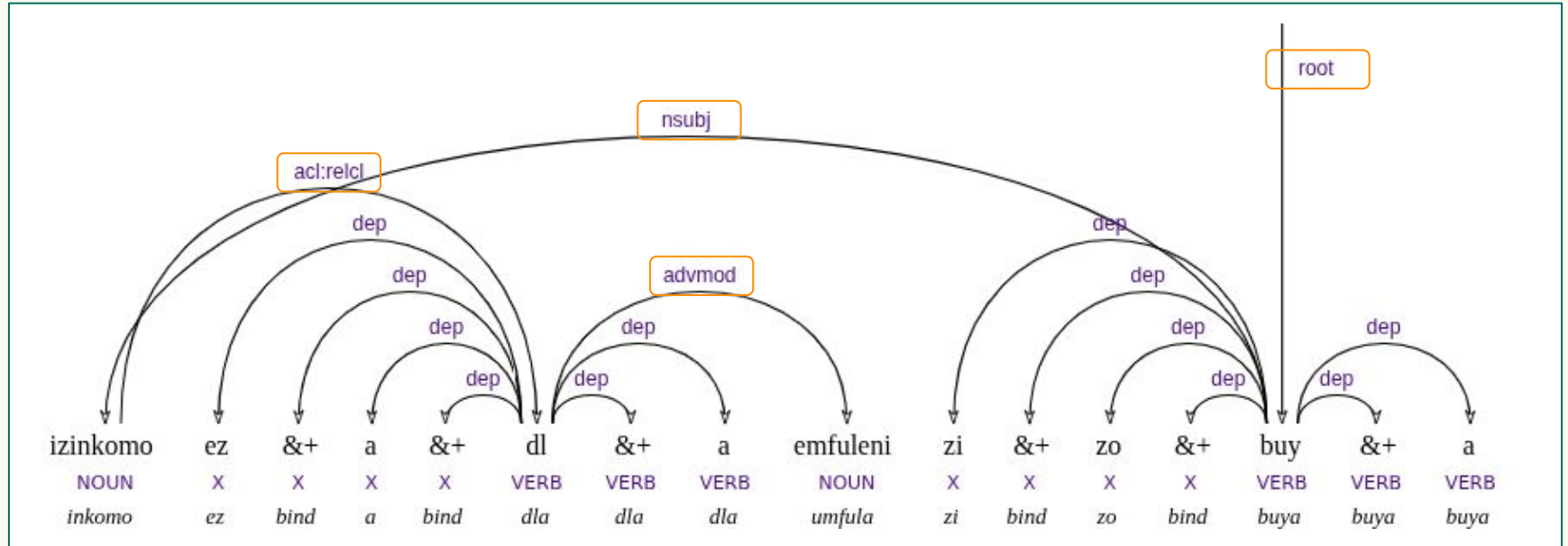
gf2ud for isiZulu

“izinkomo ezadla emfuleni zizobuya”

(the cows who ate by the river will return)

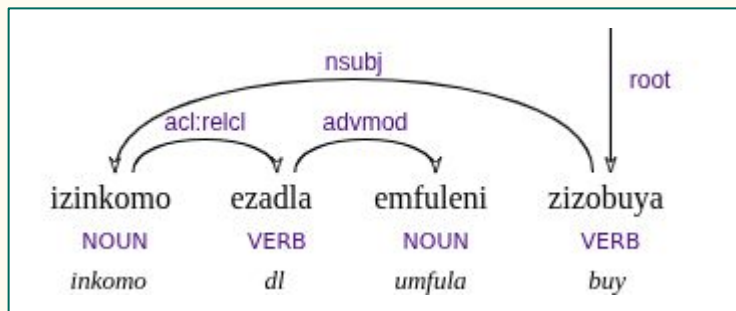


gf2ud for isiZulu

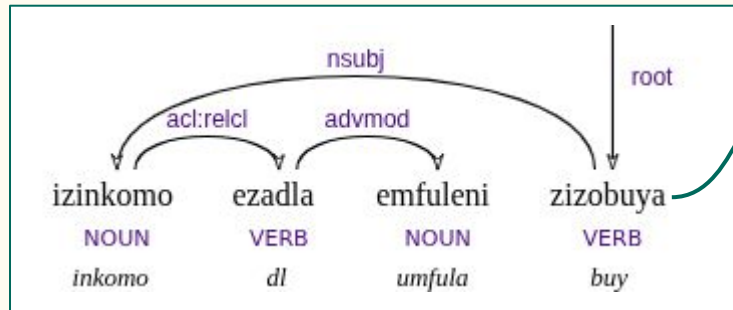


gf2ud for isiZulu

Postprocess the CoNNL-U output to consolidate tokens



gf2ud for isiZulu



Features of "zizobuya"

FORM

zizobuya

LEMMA

buy

Universal Features

+

Miscellaneous Features

+

FUN	UseV,UseV,return_V,return_V	
-----	-----------------------------	--

Edit tokens

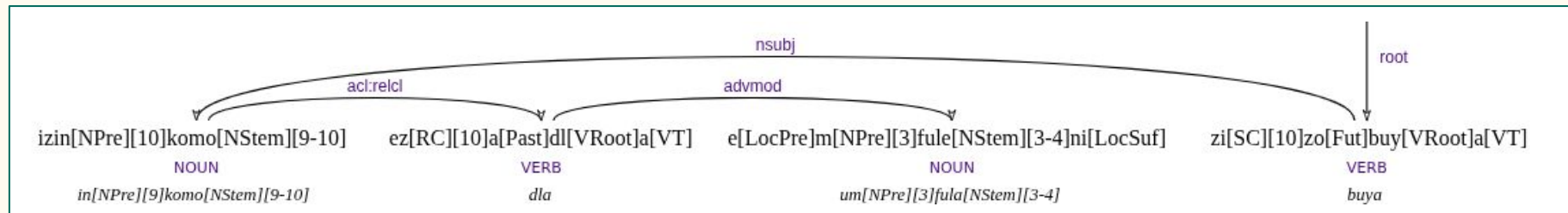
▼

CANCEL

OK

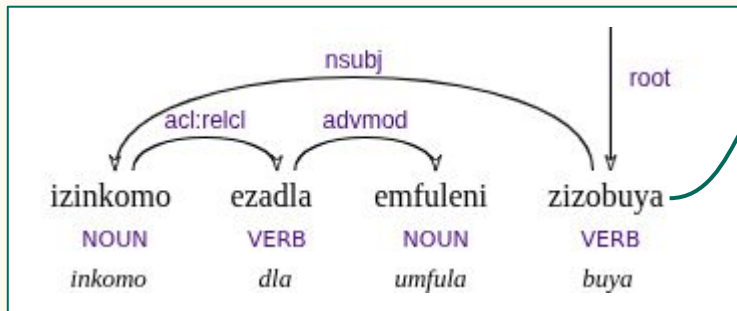
gf2ud for isiZulu

Use the morphological annotation version of the ZRG



gf2ud for isiZulu

Postprocess the CoNNL-U output to
extract features from linearisation



Features of "zizobuya" ✕

FORM
zizobuya

LEMMA
buya

Universal Features +

Polarity	Pos	✕
Tense	Fut	✕

Miscellaneous Features +

Agreement	Bantu10	✕
FUN	UseV,UseV,return_V	✕

Edit tokens ▼

CANCEL

OK

gf2ud for isiZulu

A quick look under the hood...

```
# sent_id = gfud1000019
# text = izinkomo ezadla emfuleni zizobuya
1  izinkomo    inkomo  NOUN   N      _      4      nsubj      _      FUN=cow_N|NounClass=Bantu10
2  ezadla     dla     VERB   V      Polarity=Pos|Tense=Past 1  acl:relcl  _      FUN=UseV,UseV,eat_V|Agreement=Bantu10
3  emfuleni   umfula  NOUN   N      2      advmod    FUN=river_N|NounClass=Bantu3
4  zizobuya   buya    VERB   V      Polarity=Pos|Tense=Fut 0      root      _      FUN=UseV,UseV,return_V|Agreement=Bantu10
```

Universal features

Custom features

Part-of-speech

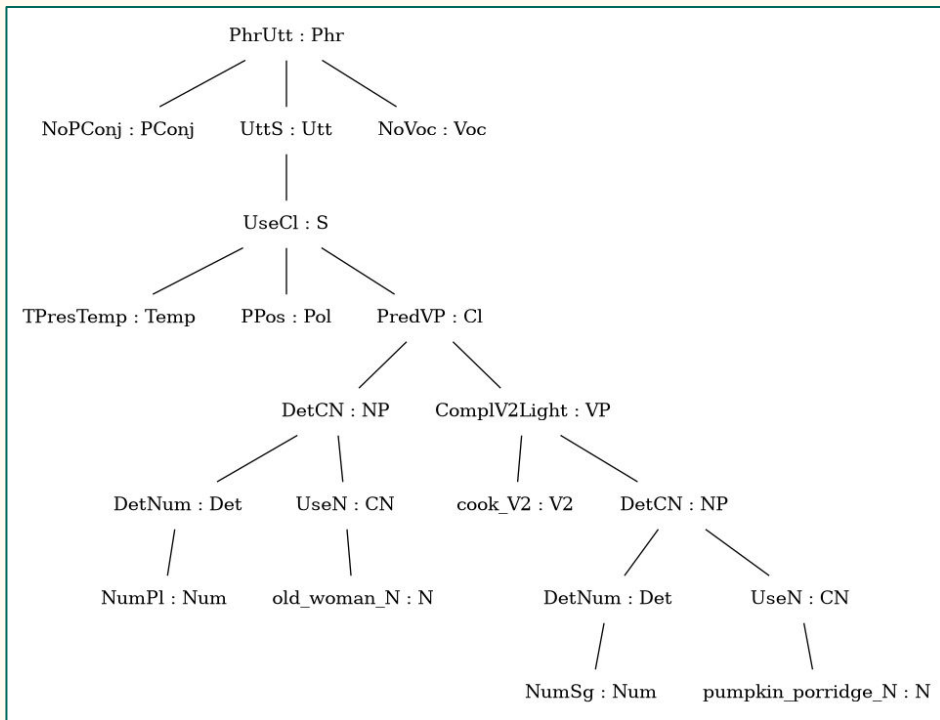
Dependency relations

gf2ud for isiZulu

Automatically convert GF treebanks to UD treebanks

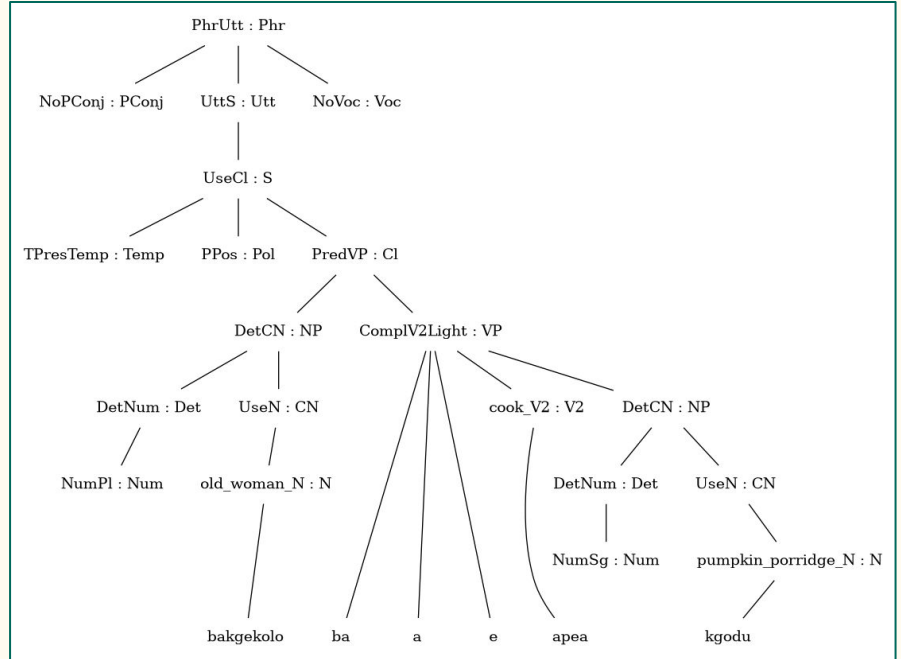
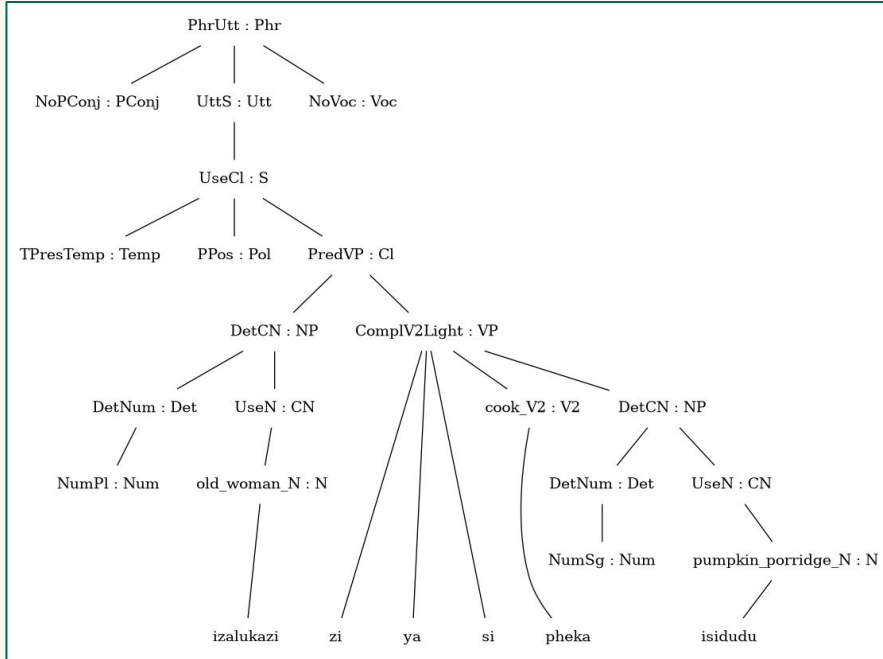
1. Obtain GF treebank
2. Define abstract annotations
3. Use morphological version of RG
4. Postprocess
 - a. consolidate tokens
 - b. extract morphological annotations
5. Obtain UD treebank!

gf2ud for Southern Bantu languages

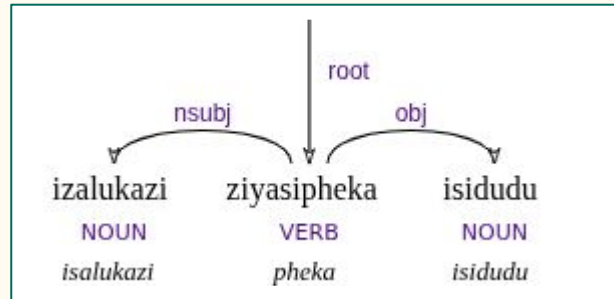


The old women will cook it, the porridge.

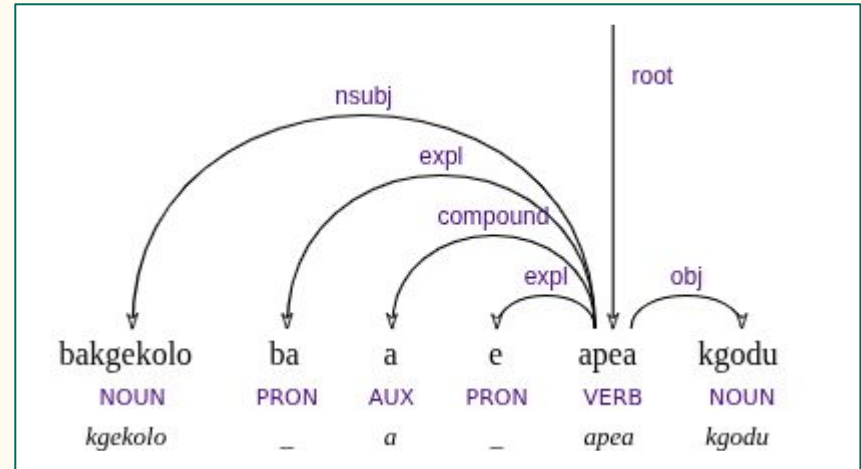
gf2ud for Southern Bantu languages



gf2ud for Southern Bantu languages



?



The old women will cook it, the porridge.

What's next?

Nginyaqonda! Foundation Phase Literacy

- Funding...
- isiXhosa, Setswana Tshivenda

Wikidata for the people of Africa

- (Wednesday)

Data annotation and augmentation

- Bootstrap morphological annotation for Siswati, isiXhosa
- Improved methods for data augmentation?

Universal Dependencies treebanks with gf2ud

- Convert larger treebank with gf2ud, train and evaluate UDPipe parser or similar
- Triangulate with Sepedi